Exercise 1

Employees

Id	First Name	Last Name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

① Write a SQL query to retrieve all columns from the employees table

```
SELECT *
FROM employees;
```

Id	First_Name	Last_Name	department	salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Fransisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Fransisco

② Write a SQL query to find all the unique departments in the employees table

```
SELECT DISTINCT departments
FROM employees;
```

Department
IT
HR
Finance
Marketing

③ Write a SQL query to retrieve all employees' first and last names, ordered by salary in descending order

```
SELECT First_Name, Last_Name
FROM employees
ORDER BY salary DESC;
```

First_Name	Last_Name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Daniel	Clark
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

④ Write a SQL query to retrieve the top 5 highest paid employees

```
SELECT *
From employees
ORDER BY salary DESC
LIMIT 5;
```

Id	First Name	Last Name	Department	Salary	hiredate	City
6	Emily	Davis	IT ~~Department~~	62000 ~~Salary~~	2015-02-14 ~~hire date~~	Chicago ~~City~~
5	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
1	John	Doe	IT ~~Finance~~	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York

⑤ Write a SQL query to find employees who work in the IT department

```
SELECT *
FROM employees
WHERE department = 'IT';
```

Id	First Name	Last_Name	department	salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-05-10	San Francisco

⑥ Write a SQL query to find employees who work in the Finance department and have a salary greater than 58000

```
SELECT *
From employees
WHERE department = 'Finance'
AND salary > 58000;
```

Id	First_Name	Last_Name	department	Salary	Hire_date	City
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Houston

⑦ Write a SQL query to find employees who work in the HR department OR the Marketing department

Id	First_Name	Last_Name	department	Salary	Hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

```
SELECT *
FROM employees
WHERE department = 'HR'
OR department = 'Marketing';
```

⑧ Write a SQL query to find employees who do NOT work in the IT department

```
SELECT *
FROM employees
WHERE department != 'IT';
```

Id	First_Name	Last_Name	department	Salary	Hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

(9) Write a SQL query to find employees who are in the HR, IT or Finance department.

```
SELECT *
FROM employees
WHERE department IN ('HR', 'IT', 'Finance');
```

Id	First Name	Last Name	department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58000	2021-03-25	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

(10) Write a SQL query to find employees who are in the IT department, have a salary greater than 50000 and are located in New York

```
SELECT*
FROM employees
WHERE department = 'IT'
AND salary > 50000
AND city = 'New York';
```

Id	First Name	Last Name	department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	58000	2021-03-25	New York

(11) Write a SQL query to retrieve the first and last names of employees who work in the Finance OR Marketing department, earn more than $2000 and order the results by salary in descending order

```
SELECT First_Name,
       Last_Name
FROM employees
WHERE department = 'Finance'
 OR department = 'Marketing'
AND salary > $2000
 ORDER BY salary DESC;
```

First_Name	Last_Name
Mike	Johnson
Robert	Wilson
Daniel	Clark

(12) Write a SQL query to find all the unique cities where employees work excluding those in the IT and HR departments

```
SELECT DISTINCT city
FROM employees
WHERE department != 'IT'
 AND department != 'HR';
```

OR

```
SELECT DISTINCT city
FROM employees
WHERE department NOT IN ('IT', 'HR');
```

City
Chicago
Los Angeles
San Francisco
Houston

(13) Write a SQL query to retrieve employees who are NOT in the Finance department, have a salary greater than 50000, and order by the hire date in ascending order

```
SELECT *
FROM employees
WHERE NOT department = 'Finance'
AND salary > 52000
ORDER BY Hire_date ASC;
```

Id	First Name	Last Name	department	Salary	City
6	Emily	Davis	IT	2015-02-14	Chicago
1	John	Doe	IT	2018-06-15	New York
4	Sarah	Brown	IT	2021-03-25	New York
9	Daniel	Clark	Marketing	2022-06-01	Chicago

(14) Write a SQL query to find the first 3 employees who work in either Chicago or Los Angeles and belong to the IT or Marketing department.

```
SELECT *
FROM employees
WHERE city IN ('Chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
limit 3;
```

Id	First Name	Last Name	Department	Salary	Hiredate	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago